Presentation 1

1. Link to presentation and code on GitHub:
    - Presentation Link: Click Here (Note: if the Power BI embedded in the slides fails, static visuals can be found at the bottom of the slide deck)
    - Coding was done on PowerBI. The data was originally transformed by my partner, Julia Schaffner, and then I built the visualization per the process below.

2. What did you do?

I created a visual for the Time Trends Dashboard using the CAR data. I built off my group partner, Julia's, work in which she compiled all CAR data and All Calls data into singular CSV files using the import code provided by Prof. Krishna. She then loaded them into PowerBI to perform more specific data transformations. Within Power BI, she used the Group By and Advanced Group By functions to ensure each Contact Session ID in the CAR dataset appeared only once, preventing double counting from multiple actions per session. Julia then created four new aggregated columns—Count, Call Start Time, Starting Hour, and All Rows—and expanded the nested tables to access detailed session information in the visuals.

With this data, I created a line chart visualization that aimed to examine if there was a seasonal peak in the number of calls received by Legal Aid. I made sure to filter the data so it only displays call volumes from 8/1/2024 - 7/31/2025 to ensure no double counting. The X-axis is the month in which the call took place over the course of this calendar year. The Y-axis is the number of unique calls during that month. On each visual, you can filter by EP Name, Flow Name, Termination Reason, and the axes for the visual.

3. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it currently looks like call volumes acutely spike in the new year, dip moderately during the summer months, and drop off dramatically in December. With this in mind, Legal Aid could leverage short-term staff or AI-powered agents during the new year while cutting back during the summer and the holidays. By aligning staffing and resources with periods of highest demand, the organization can serve more clients effectively without increasing costs, improving both accessibility and overall system performance.

4. Issues faced, attempt to resolve issues, and resolved issues:

The major issue I, along with the rest of my group, are facing is that the data has a lot of confusing and/or missing terms and information. We were able to get some context about acronyms and terms used by searching the internet or referencing the document Professor Arvind uploaded to GitHub; however, we still have questions about how the different answers in a column compare to each other in context of the overall business problem (i.e. SIP vs WXCC vs. WXC).

5. Next Steps

    - Drill down further to see if the call volume trend observed above is uniform across all menu types or if specific menu options follow unique trends and thus require different solutions/recommendations

- Compare if the seasonal spike from April to September is true in other years suggesting a regular trend or if it is more closely related to the extreme policy changes from this summer; to do so, I would create this same visualization with data from years past
- Refine the dashboard visuals to clearly display trends in peak call times by current filters; to do so, I would create heatmaps or bar graphs to highlight patterns across hours and days
- Add more filtering options to allow deeper analysis of when and why specific termination reasons happen
- Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

6. References

- Input code from Krish
- Julia's data transformations in python and PowerBI
- Information from Cynthia during class with her

Presentation 2

1. Link to presentation and code on GitHub:
    - Presentation Link: Click Here (Note: static visuals used in presentation; to see live dashboard use link below)
    - Dashboard Link: Click Here
    - Data cleaning code: Click Here

2. What did you do?

This week, my goal was to build off my previous visualization—which tracks call volumes by month—by adding hierarchical filtering by year and month. This way, Cynthia can easily compare this trend across the relevant time frames that she needs. To do so, I had to first download the new CAR data for the month of September. Then, building off of Krish's input code, I added a couple lines that ensure that we are only counting unique session IDs (similar to Julia's code). After this, I created a column that tracks the month of the call and a column that tracks the year of the call; these will be helpful when creating the hierarchical filter.

Next, I downloaded this cleaned monthly as a single .CSV and uploaded it to Power BI. Here, I created a line plot with the months in a calendar year as the x-axis and unique call session ID as the y-axis. I then added a hierarchical filter that easily allows the viewer to choose a specific year (either 2024 or 2025) and hone on a specific month(s).

While there did seem to be a slight increase in calls during summer 2024 and during the near year 2025, overall there did not appear to be any substantial seasonal trend that occurred in both years. However, I did notice that there have been less calls per month for most of 2025 compared to 2024. On average, it looks like about one to two thousand less calls per month, which would add up to twelve to twenty-four thousand calls a year. Though I have not been able to make any concrete recommendations based on this, I have a hypothesis moving forward that I want to test: if this drop off can be accounted for by certain issues being either resolved or more likely people feel comfortable reaching out for help due to today's political and social climate.

Moving forward, I hope to understand why there was this call drop off and if certain EP names (i.e. immigration, HIV, etc.) are responsible for this.

3. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it currently looks like call volumes on track to be less for 2025 compared to 2024. We need to understand why this is occurring before making any concrete recommendations. However, I wonder if certain issues might be best served by other resources, which would free up agents so that the organization can serve more clients effectively without significantly increasing costs.

4. Issues faced, attempt to resolve issues, and resolved issues:

One issue I faced is that when I was originally trying to add a filter for EP name, there ended up only being three potential EP name options post-data cleaning. This does not seem right so it must be an issue related to how I cleaned my data. While I tried to resolve it and build out a visualization for this week, I was not able to get there. This upcoming week I will try and do so.

5. Next Steps

- Create new visualization to understand why the drop off in calls occurred in 2025; this would plot call volume total for each EP name against months of the year with the goal of understanding which reason for calling has declined the most
- Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

6. References

- Input code from Krish
- Julia’s data transformations in python and PowerBI
- Information from Cynthia during class with her